QP Code: 25901159

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER Value Addition Course (VAC) –

MG4VACCHE200 - BASIC ENVIRONMENTAL CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes Maximum Marks: 50

Part A – Multiple Choice Questions (1 mark each)

1. The ozone layer is found in Ans: b) Stratosphere
2. Identify the technological factor that has contributed to increase global warming.
   Ans : b) Industrialization using fossil fuels
3. Without the natural green house effect, Earth's average temperature would be approximately:
   Ans: b) −18°C
4. Excess anthropogenic greenhouse gases lead to: Ans : b) Global warming
5. Avoiding open burning of waste helps in: Ans: b) Reducing air pollution
6. The First stage of wastewater treatment: Ans: c) Primary treatment
7. The Main product formed when hydrogen is used as fuel: Ans: c) Water
8. Which Practice supports sustainable mineral utilization: Ans: c) Metal recycling
9. Ethanol from biomass produced by: Ans: b) Fermentation of sugars

Part B – Short Answer Questions (Any 5 × 4 = 20 marks)

10 How does ozone behave as a green house gas?

Ozone present in the atmosphere absorbs infrared radiation emitted from the Earth's surface. By absorbing this radiation it traps heat within the atmosphere and contributes to the greenhouse effect. This trapped heat increases the temperature of the lower atmosphere and influences climate regulation. Although ozone is essential for absorbing harmful ultraviolet radiation in the stratosphere, increased ozone in the lower atmosphere can contribute to atmospheric warming. (Any four valid points, give full credit)

11. What is sequestration of CO_2?

Carbon dioxide sequestration refers to the capture and long-term storage of carbon dioxide from the atmosphere or from industrial emission sources in order to reduce greenhouse gas concentration. The captured CO_2 is prevented from entering the atmosphere and is stored in geological formations, deep saline aquifers, depleted oil and gas reservoirs, or through mineral carbonation processes. This technology is an important strategy for mitigating climate change because it reduces the accumulation of carbon dioxide responsible for global warming. (Definition – 2 marks, methods of storage or role in climate mitigation –2 marks).

12. What is ozone depletion? Mention any two causes of ozone layer depletion.

Ozone depletion refers to the thinning or reduction in the concentration of the ozone layer in the stratosphere. This occurs mainly due to the release of chemical substances that destroy ozone molecules through catalytic reactions. Chlorofluorocarbons (CFCs) used in refrigerants and aerosols release chlorine radicals in the atmosphere which break down ozone molecules. Other pollutants such

as nitrogen oxides, halons and certain industrial chemicals also contribute to the destruction of the ozone layer. (Definition – 2 marks, two causes – 2 marks)

13. What is acid rain? Mention any two causes of acid rain?

Acid rain refers to precipitation such as rain, snow or fog that contains acidic components with a pH lower than about 5.6. It is mainly produced when sulfur dioxide (SO_2) and nitrogen oxides (NO_x) released from industrial processes and vehicle emissions react with water vapour in the atmosphere to form sulfuric acid and nitric acid. These acids dissolve in rainwater and fall to the Earth's surface, causing harmful effects on soil, vegetation, aquatic systems and monuments. (Definition – 2 marks, two causes – 2 marks).

14. List four benefits of biomass utilization in decarbonization.

Biomass utilization involves the use of organic materials such as agricultural residues, wood waste and plant materials as energy sources. Biomass is considered renewable because plants absorb carbon dioxide during growth, which offsets the CO_2 released during energy production. The use of biomass reduces dependence on fossil fuels and helps decrease net greenhouse gas emissions. It also supports waste management by converting organic waste into useful energy resources. (Any four valid points give full credit).

15. List four advantages of solar energy.

Solar energy is one of the most abundant and renewable sources of energy available on Earth. It produces electricity without emitting greenhouse gases or air pollutants during operation, making it environmentally friendly. Solar energy systems reduce dependence on fossil fuels and contribute to sustainable energy development. Once installed, solar power systems have relatively low operating and maintenance costs, making them economically beneficial in the long term. (Any four advantages, give full credit).

16. Define mining. Discuss the purpose of mining.

Mining is the process of extracting valuable minerals and geological materials from the Earth's crust. These minerals include metals such as iron, copper and aluminium as well as fuels and industrial materials. Mining provides essential raw materials required for industries, manufacturing, construction and technological development. It also plays an important role in economic growth by supporting industrial production and infrastructure development. (Definition – 1 mark, purpose and economic importance – 3 marks).

Part C – Short Essay (Any 3 × 7 = 21 marks)

17. Explain the characteristics of thermosphere and exosphere.

The thermosphere is the layer of the atmosphere located above the mesosphere and extends roughly from about 80 km to nearly 700 km above the Earth's surface. In this region the temperature increases significantly due to the absorption of high-energy solar radiation. The thermosphere contains the ionosphere, which is rich in charged particles and plays an important role in radio communication and satellite transmission. The density of air molecules is extremely low in this layer. Above the thermosphere lies the exosphere, which is the outermost layer of the atmosphere. In the exosphere, atmospheric gases gradually escape into outer space and the density of particles becomes extremely sparse. Hydrogen and helium are the dominant gases present and many artificial satellites orbit within

this region. (Thermosphere definition – 1 mark, characteristics – 2.5 marks, exosphere definition – 1, characteristics – 2.5 marks,).

18. Explain in detail the process of removing CO_2 from the atmosphere using Direct Air Capture technology.

Direct Air Capture (DAC) is an advanced technology designed to remove carbon dioxide directly from the ambient atmosphere. In this process, large volumes of air are passed through specialized chemical filters or sorbent materials that selectively bind with carbon dioxide molecules. Once the CO_2 is captured, the sorbent is heated or chemically treated to release concentrated carbon dioxide gas. The captured CO_2 is then compressed and transported for long-term storage in geological formations or used in industrial applications such as synthetic fuel production. DAC technology is considered an important tool for reducing atmospheric greenhouse gas concentrations and combating climate change. (Any four valid points, give full credit).

19. Discuss in detail the major causes of water pollution. Classify the different categories of water pollution and explain each with suitable examples.

Water pollution refers to the contamination of water bodies such as rivers, lakes and groundwater by harmful substances that degrade water quality and make it unsafe for human use and aquatic life. Major causes include the discharge of untreated industrial effluents containing heavy metals and toxic chemicals, domestic sewage that introduces pathogens and organic waste and agricultural runoff carrying fertilizers and pesticides. Water pollution can be categorized into biological pollution caused by microorganisms, chemical pollution caused by toxic compounds and thermal pollution caused by the discharge of heated water from industrial plants. These pollutants disrupt aquatic ecosystems and affect human health. (Causes 2 marks, classification 3 marks explanation with examples 2 marks).

20. Explain biomass energy and discuss various biofuels obtained from biomass.

Biomass energy refers to energy obtained from organic materials such as plant residues, agricultural waste, wood and animal waste. These materials can be converted into useful fuels through processes such as combustion, fermentation and anaerobic digestion. Common biofuels derived from biomass include bioethanol produced by fermentation of sugars from crops such as sugarcane or corn, biodiesel produced from vegetable oils or animal fats and biogas produced from the anaerobic digestion of organic waste. Biomass energy is considered renewable and helps reduce reliance on fossil fuels while contributing to sustainable energy production. (Definition – 1, any three biofuel with explanation 6 marks).

21. Discuss metallic and non-metallic mineral resources in detail, with examples and properties.

Mineral resources are naturally occurring inorganic substances found in the Earth's crust that are economically valuable. Metallic minerals contain metal elements and are good conductors of heat and electricity; examples include iron, copper and aluminium ores. These minerals are widely used in construction, manufacturing and electrical industries due to properties such as malleability and ductility. Non-metallic minerals do not contain metals and generally lack metallic properties. Examples include limestone, gypsum, mica and phosphates, which are used in cement production, fertilizers, ceramics and other industrial applications. (Definition – 1, metallic minerals examples – 3marks, non-metallic minerals examples –3 marks).